

CSC422 Fall 2025 - Project Proposal

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1 Chosen Project

Title: Time-series Forecasting of Energy Consumption

Category: **Predictive Modeling**

Dataset: PJM Hourly Energy Consumption Dataset (Kaggle)

We will be using the PJME region data from this dataset, although the dataset spans 20 years the regions included change over this period. Additionally, chronological separation of this dataset across regions would lead to different region makeup at the beginning and end of this period.

<https://www.kaggle.com/datasets/robikscube/hourly-energy-consumption/data>

Difficulty: ★★★★★

2 Project Abstract

Electricity consumption varies by many factors. Some examples include the hour of the day, day of the week, or whether it is a holiday. For power grid operators, predicting the usage accurately is important for being able to scale production of electricity up or down to meet needs and plan staffing. Consumer demand is the primary driver of electricity demand. Consumer demand itself is subject to many factors, including the previous factors, as well as the need for air conditioning and heating (dependent on local weather conditions). In this project, we will analyze the PJME regional data series from the PJM hourly energy consumption dataset. We will develop and train two models to predict short term electricity demand using features constructed from the original dataset of only usage and time.

Our first analysis will be dedicated to finding patterns in the historical data to guide our feature selection and engineering. Before this analysis we will expand the simple time and usage attributes to add features like the day of the week, week of the month, month of the year, season/quarter, holiday, and other features that we believe may add predictive value like the local weather conditions in the region to estimate the need for air conditioning or heating. We may also include features for longer term consumption patterns showing trends of increasing or decreasing consumption in recent history. Weather will be an outside source of information and will be included from a source like NOAA in the feature set, if practical to clean and organize. Once the attributes/features are expanded from the initial dataset, we will use methods to find which are likely to have significant predictive power using Principal Component Analysis (PCA) or potentially Non-Negative Matrix Factorization (NNMF) if no attribute contains negative values. This will give us our first glimpse of feature importance in the training of our decision tree-based models and inform the maximum depth of the trees in the models we train.

After this exploration, we will create several models, a simple moving average forecast that will form a baseline for measuring the improvement that the later models add. We will then build a decision tree model and random forest model to train on our feature set. Both tree based models will run from the same data pipeline.

Our objective will be to evaluate the different modeling approaches to find which offers the best predictions. By using decision-tree based models we plan to explain feature importance and compare the results of our exploratory data analysis with PCA/NNMF to the feature importances found by our tree based models.

3 Proposed Methodology

This project will primarily be implemented in Python utilizing several applicable libraries: Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, and Statsmodels. Development will be done in VS Code and Jupyter Notebooks for data analysis.

We will first need to extract the PJME series from the PJM hourly energy consumption dataset. Preprocessing for this data will first be concerned with addressing missing data, duplicate data, inconsistent types, and outliers. Time stamps for the hourly usage will then be expanded into a consistent datetime format.

The feature set will be expanded from the initial timestamp and usage to include these additional features:

- Hour of the day
- Day of the week
- Month
- Year
- Weekend/weekday
- Season/quarter
- Holiday Category (0 = stay-inholiday, 1 = regular day, 2 = travel holiday, research to classify)(We will attempt to make this ordinal)
- Lagged energy consumption (trend short, medium, and/or long)
- Rolling averages of previous Consumption
- Weather conditions in relation to population (NOAA calculates as the difference of the local temperature from 65F and weighted by population)

We will visualize the data using Matplotlib and/or Seaborn to get a better understanding of the correlations and distributions of the feature set. Additionally, by exploring the dataset with PCA/NNMF, we will attempt to identify a precursory understanding feature-importance which will help in informing the maximum tree depth when training for our tree based models. NNMF will be used if features do not contain negative values, PCA will be used regardless.

The data will be organized chronologically and separated into train and test periods for the PJME region series. The separations for train and test will follow this chronological order with the test set period coming after the training set. This arrangement will prevent future leaks, and embargo range will be used if needed to make separations in data overlap that might be caused

by trend or moving average features. Only using the PJME dataset from the larger series will prevent the uneven regional representation across the datasets' time range.

The baseline that we will compare the tree models to will be a simple moving average offering a simple prediction based on recent past usage. The two starter models planned are a decision tree and random forest model. Both models will share the same preprocessing steps and the maximum tree depth will be informed by the results of the exploratory analysis and the preliminary feature importance. An expansion into a gradient boosted or neural network model may be considered if time allows.

Model performance will be evaluated with these metrics:

- **Root Mean Squared Error (RMSE)**
- **Mean Absolute Percentage Error (MAPE)**
- **Mean Absolute Error (MAE)**

These metrics will be evaluated to determine which model generates the most accurate predictions for electricity usage. In the final report, we will compare the performance of the baseline simple moving average, the decision tree model, and random forest. We will also compare the feature importances of the models and compare these to the predictions of feature importance derived by the exploratory analysis using PCA and/or NNMF.

4 Expected Outcomes

From the expected outcomes, we will identify several distinct patterns within the electricity consumption data set from PJM. We expect to see energy consumption vary considerably depending on a multitude of factors. These expected factors include the time of day, as certain times will show higher demand in both residential and commercial activities vs those when activity is lower. There is also an expectation that the weekends and weekdays will differ given the change from workday vs personal activities will draw different usage patterns.

Space heating is by far the largest energy consumer in a residential home. As such, a major factor of consumption we are expecting to see is the change of seasons. Electricity usage could see an increase in times of higher temperature disparities from neutral temps. Winter and summer will be the times where electricity usage is expected to be the highest. The colder average temperatures from the winter will cause more electricity usage for heating, while summer will have more people out of school trying to stay cool during the hotter weather.

The last possible external factor that we are looking at that could affect the usage rates of electricity are holidays. Holidays have the possibility to drastically change the population in the selected areas

along with nominal patterns of life. This effect could also stretch around the holiday, from travel to more production rates in factories leading to higher drawing of commercial power.

We expect that long-term visualizations should reveal broader trends and changes within electricity consumption over the years from the given dataset. From analyzing this we expect to append to the list of factors or hypothesize further possible factors.

For the predictive modeling aspect of the project, there is an expectation that a basic baseline forecast will prove less effective than advanced models incorporating historical energy data alongside time-centric attributes. Two key factors that we expect to significantly enhance predictive capabilities are lagged consumption values along with recurring daily and weekly usage patterns.

The final deliverables for this project will be the following ten topics.

- 1) Cleaned PJME dataset along with a reproducible preprocessing pipeline
- 2) Centralized metadata repository corresponding with the data set
- 3) Visualization of all listed factors explored
- 4) Non-negative matrix factorization feature-loading analysis (possibly find better wording)
- 5) Moving average baseline results
- 6) Decision Tree and Random Forest models
- 7) RMSE, MAPE, and MAE comparison plots vs actual results
- 8) Feature importance comparisons with visualizations
- 9) Complete repository and instructions
- 10) Final written report and presentation

These topics will be assigned to member(s) of the team at the start of the implementation. The assigned member(s) are subject to change during the implementation if necessary.

5 Timeline

Week 1: Collect the data and explore it with the PJM dataset and begin preprocessing.

Firstly, download the PJM hourly energy-consumption dataset and then restrict the working data to the PJME region. This data will all come from Kaggle. Then, everyone will go through the entire set and fully understand the dataset structure, variables, and time range. Afterward, members will perform different visual tests, make sure the dataset is complete, and try to spot potential problems we could run into. If applicable, we will collate data from outside weather sources, such as NOAA, as an extra feature to train our models with later.

Week 2: Clean the data and complete preprocessing.

Firstly, the group will convert all the timestamps into a consistent datetime format. It will be a lot easier to work with if it all has the same format and style. From last week, we will have identified missing values and/or duplicates, and here we will need to work on fixing those. Next, members will generate time-based features, categories, and data pruning. These are all parts of data preprocessing that we learned about in class, and the group will implement those. Lastly, everyone

will need to thoroughly check and organize the data for analysis.

Week 3: Explore the Data and Perform Analysis

In this stage, we will continue to analyze hourly, daily, weekly, and monthly energy-consumption patterns. Using just visuals, we will see if we can find patterns even before implementing the machine learning model. Then, members will investigate all the differences between weekdays and weekends. We want to see if day and time make a difference. We will then compare consumption across regular days, major holidays, and minor holidays. Holidays are a good indicator for energy spikes because everyone stays home. We will also then apply NMF to the nonnegative engineered feature matrix and look at the resulting feature loadings.

Week 4: Build the shared feature pipeline and baseline models

Firstly, we will build and then implement persistence and moving-average baseline forecasts. A Decision Tree model using the shared engineered feature matrix that was built last week will also be implemented, tested, and used during this step. We will then create chronological PJME training and testing datasets. Finally, once everything is working properly, all members must individually evaluate initial model performance. Then, they will report back to see if there are any inconsistencies and ensure everything is working well.

Week 5: Develop and implement the two Tree-Based models

Firstly, the team together will make a Random Forest model using the same feature matrix as the Decision Tree. The two models will be made in order to do some comparison when we actually run our results. Make sure to incorporate lagged and rolling-average features. Now, we will actually run the tests and compare the Decision Tree and Random Forest; attempting Gradient Boosting could potentially lead to better results. If something does not go our way during the test, we will tune the forest and make sure it is properly aligned with the dataset.

Week 6: Look at the numbers, evaluate the model, and perform all comparison runs

Firstly, the team must look at the baseline numbers because they perform surprisingly well. If the models do not beat the baseline, then some tuning or redoing must happen. To evaluate the baseline, Decision Tree, and Random Forest, we will be using MAE, RMSE, and MAPE. We will compare all the different model predictions with actual electricity demand as well. Some forecasting models perform well in areas of the dataset and poorly in other areas. Lastly, we will select the best-performing approach.

Week 7: Perform a thorough and complete final analysis as well as a visualization

To showcase the numbers, the team will create final graphs and tables. It is crucial to show the differences and visualize the entire experiment because we can see what models perform well and where others do not. The team will interpret the most important energy-consumption patterns. Then, we will compare NMF feature loadings with Decision Tree and Random Forest feature importances to see how different the numbers are. Finally, we will need to create a nice narrative storyline in order to put our building, experiment, and results into a coherent timeline.

Week 8: Make the entire final report and present

Assuming the team will present both models and the experiment in class, then we will make Google Slides. This, of course, depends on whether or not we actually showcase our project; however, having slides just in case gives a good refresher and verification of our output. Then, the team will complete the written project report. We will prepare all the different final visualizations. Finally, we will finalize conclusions and project deliverables.

6 Possible Challenges

The team evaluated all the different possible challenges that could happen when deciding how to build and implement the machine learning models. Some problems could be completely out of our control and have to do with the Kaggle dataset. The size of the entire hourly PJM dataset is probably the biggest bottleneck and problem we have in this experiment. To address this issue, we will reduce both computation and inconsistencies by using only the PJME series. Pandas functions will also help increase project efficiency.

Energy consumption is affected by all sorts of different external factors. The temperature during hot summer peaks and cold winter spikes, as well as severe weather, could make energy consumption difficult to predict. When evaluating the table, we will look for all the different timestamped weather series that can be aligned with PJME. If there is no series available, then we will document weather as a limitation. In addition to weather anomalies, holidays have a profound impact on the energy market due to an increase of families staying home. This is especially true during major holidays like Christmas and New Year's. The team will classify observations as regular days (0), major holidays (1), or minor holidays (2). We will compare each holiday group with typical consumption for similar-looking periods. Lastly, when we start our comparisons between the Decision Tree and the Random Forest, both models must remain achievable and fit within the project schedule. Our hope is to make at least one of those two models perform better than the baseline forecast benchmarks. Otherwise, the use of machine learning in this experiment would be debunked.

Another procedure we can employ to improve our results are gradient boosting and researching different library functions. This extension can be performed after the required deliverables are complete.